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Method and apparatus for receiving MAC address of another STA within reception MLD in wireless LAN system

Abstract

Proposed are a method and an apparatus for receiving an MAC address of another STA within a reception MLD in a wireless LAN system. Specifically, the reception MLD receives an ML element from a transmission MLD through a first link. The reception MLD decodes the ML element. The ML element includes common information and link information. The link information includes a profile field of the second transmission STA. The profile field of the second transmission STA includes first information relating to whether an MAC address of the second transmission STA exists. If the first information is configured to be 1, the profile field of the second transmission STA includes the MAC address of the second transmission STA.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)
(1) This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2022/000022, filed on Jan. 3, 2022, which claims the benefit of earlier filing date and right of priority to Korean Application Nos. 10-2021-0002733, filed on Jan. 8, 2021, and 10-2021-0006183, filed on Jan. 15, 2021, the contents of which are all incorporated by reference herein in their entirety.
TECHNICAL FIELD
(2) The present specification relates a multi-link operation in a wireless local area network (WLAN) system and, most particularly, to a method and apparatus for receiving a MAC address of another STA in a receiving MLD.
BACKGROUND
(3) A wireless local area network (WLAN) has been improved in various ways. For example, the IEEE 802.11ax standard proposed an improved communication environment using orthogonal frequency division multiple access (OFDMA) and downlink multi-user multiple input multiple output (DL MU MIMO) techniques.
(4) The present specification proposes a technical feature that can be utilized in a new communication standard. For example, the new communication standard may be an extreme high throughput (EHT) standard which is currently being discussed. The EHT standard may use an increased bandwidth, an enhanced PHY layer protocol data unit (PPDU) structure, an enhanced sequence, a hybrid automatic repeat request (HARQ) scheme, or the like, which is newly proposed. The EHT standard may be called the IEEE 802.11be standard.
(5) In a new WLAN standard, an increased number of spatial streams may be used. In this case, in order to properly use the increased number of spatial streams, a signaling technique in the WLAN system may need to be improved.
SUMMARY
(6) The present specification proposes a method and apparatus for receiving a MAC address of another STA in a receiving MLD in a wireless LAN system.
(7) An example of this specification proposes a method for receiving the MAC address of another STA in the receiving MLD.
(8) The present embodiment may be performed in a network environment in which a next generation WLAN system (IEEE 802.11be or EHT WLAN system) is supported. The next generation wireless LAN system is a WLAN system that is enhanced from an 802.11ax system and

may, therefore, satisfy backward compatibility with the 802.11ax system.
(9) The present embodiment proposes a method and apparatus for receiving, by a non-AP STA, a MAC address of another non-AP STA in the same non-AP MLD through an ML element in MLD communication. The non-AP STA recognizes another non-AP STA based on the MAC address of the other non-AP STA, so that frame exchange is possible on a corresponding link. Here, the transmitting MLD may correspond to an AP MLD, and the receiving MLD may correspond to a non-AP MLD. If the non-AP STA is a first receiving STA, a first transmitting STA connected to the first receiving STA through a first link may be referred to as a peer AP, the second to third transmitting STAs connected through different links may be referred to as other APs, and the second and third receiving STAs connected through different links may be referred to as other non-AP STAs.
(10) A receiving Multi-link Device (MLD) receives a multi-link (ML) element from a transmitting MLD through a first link.
(11) The receiving MLD decodes the ML element.
(12) The transmitting MLD includes a first transmitting station (STA) operating on the first link and a second transmitting STA operating on a second link. The receiving MLD may include a first receiving STA operating on the first link and a second receiving STA operating on the second link.
(13) The ML element includes common information and link information.
(14) The link information includes a profile field of the second transmitting STA. The profile field of the second transmitting STA includes first information on whether a MAC address of the second transmitting STA is present. If the first information is set to 1, the profile field of the second transmitting STA includes the MAC address of the second transmitting STA. When the first information is set to 0, the profile field of the second transmitting STA may not include the MAC address of the second transmitting STA.
(15) The transmitting MLD may further include a third transmitting STA operating in a third link. The receiving MLD may further include a third transmitting STA operating in the third link.
(16) The link information may further include a profile field of the third transmitting STA. The profile field of the third transmitting STA may include second information on whether a MAC address of the third transmitting STA is present. When the second information is set to 1, the profile field of the third transmitting STA may include the MAC address of the third transmitting STA. When the second information is set to 0, the profile field of the third transmitting STA may not include the MAC address of the third transmitting STA.
(17) The first link may be an association link, and the second and third links may be non-association links.
(18) According to the embodiment proposed in this specification, by receiving the MAC address of an STA operating in a non-associated link through an associated link, there is an effect of enabling frame exchange in the non-associated link.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows an example of a transmitting apparatus and/or receiving apparatus of the present specification.
(2) FIG. **2** is a conceptual view illustrating the structure of a wireless local area network (WLAN).
(3) FIG. **3** illustrates a general link setup process.
(4) FIG. **4** illustrates an example of a PPDU used in an IEEE standard.
(5) FIG. **5** illustrates an operation based on UL-MU.
(6) FIG. **6** illustrates an example of a trigger frame.
(7) FIG. **7** illustrates an example of a common information field of a trigger frame.

(8) FIG. **8** illustrates an example of a subfield included in a per user information field.
(9) FIG. **9** describes a technical feature of the UORA scheme.
(10) FIG. **10** illustrates an example of a PPDU used in the present specification.
(11) FIG. **11** illustrates an example of a modified transmission device and/or receiving device of the present specification.
(12) FIG. **12** shows an example of a structure of a STA MLD.
(13) FIG. **13** shows an example of setting Multi-Link.
(14) FIG. **14** shows an example of an MLD Per-STA MAC address field.
(15) FIG. **15** shows an example of an MLD Per-STA MAC address element.
(16) FIG. **16** illustrates structures of a Multi-link Control field and a Common Info field of a Multi-link Element.
(17) FIG. **17** shows the structure of the Link Info field of Multi-link Element.
(18) FIG. **18** shows an example in which the MLD Per-STA MAC address field is included in the Common Info field of the Multi-link Element.
(19) FIG. **19** shows an example in which the MAC address of the STA is included in the Per-STA Profile subfield of the Link Info field of the Multi-link Element.
(20) FIG. **20** shows an example in which the Complete Profile is included in the Per-STA Profile subfield of the Link Info field of the Multi-link Element.
(21) FIG. **21** shows an example of including MAC address present in the Per-STA Profile subfield of the Link Info field of the multi-link element.
(22) FIG. **22** shows an example of transmitting an Initial frame on a non-association link.
(23) FIG. **23** shows another example of transmitting an Initial frame on a non-association link.
(24) FIG. **24** shows another example of transmitting an Initial frame on a non-association link.
(25) FIG. **25** shows another example of transmitting an Initial frame on a non-association link.
(26) FIG. **26** shows another example of transmitting an Initial frame on a non-association link.
(27) FIG. **27** shows another example of transmitting an Initial frame on a non-association link.
(28) FIG. **28** is a flowchart illustrating a procedure in which a transmitting MLD provides MAC addresses of other STAs to a receiving MLD through a Multi-Link element (ML element) according to the present embodiment.
(29) FIG. **29** is a flowchart illustrating a procedure in which a receiving MLD receives MAC addresses of other STAs from a transmitting MLD through a Multi-Link element (ML element) according to the present embodiment.

DETAILED DESCRIPTION

(30) In the present specification, “A or B” may mean “only A”, “only B” or “both A and B”. In other words, in the present specification, “A or B” may be interpreted as “A and/or B”. For example, in the present specification, “A, B, or C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, C”.
(31) A slash (/) or comma used in the present specification may mean “and/or”. For example, “A/B” may mean “A and/or B”. Accordingly, “A/B” may mean “only A”, “only B”, or “both A and B”. For example, “A, B, C” may mean “A, B, or C”.
(32) In the present specification, “at least one of A and B” may mean “only A”, “only B”, or “both A and B”. In addition, in the present specification, the expression “at least one of A or B” or “at least one of A and/or B” may be interpreted as “at least one of A and B”.
(33) In addition, in the present specification, “at least one of A, B, and C” may mean “only A”, “only B”, “only C”, or “any combination of A, B, and C”. In addition, “at least one of A, B, or C” or “at least one of A, B, and/or C” may mean “at least one of A, B, and C”.
(34) In addition, a parenthesis used in the present specification may mean “for example”. Specifically, when indicated as “control information (EHT-signal)”, it may denote that “EHT-signal” is proposed as an example of the “control information”. In other words, the “control information” of the present specification is not limited to “EHT-signal”, and “EHT-signal” may be

proposed as an example of the “control information”. In addition, when indicated as “control information (i.e., EHT-signal)”, it may also mean that “EHT-signal” is proposed as an example of the “control information”.
(35) Technical features described individually in one figure in the present specification may be individually implemented, or may be simultaneously implemented.
(36) The following example of the present specification may be applied to various wireless communication systems. For example, the following example of the present specification may be applied to a wireless local area network (WLAN) system. For example, the present specification may be applied to the IEEE 802.11a/g/n/ac standard or the IEEE 802.11ax standard. In addition, the present specification may also be applied to the newly proposed EHT standard or IEEE 802.11be standard. In addition, the example of the present specification may also be applied to a new WLAN standard enhanced from the EHT standard or the IEEE 802.11be standard. In addition, the example of the present specification may be applied to a mobile communication system. For example, it may be applied to a mobile communication system based on long term evolution (LTE) depending on a 3.sup.rd generation partnership project (3GPP) standard and based on evolution of the LTE. In addition, the example of the present specification may be applied to a communication system of a 5G NR standard based on the 3GPP standard.
(37) Hereinafter, in order to describe a technical feature of the present specification, a technical feature applicable to the present specification will be described.
(38) FIG. **1** shows an example of a transmitting apparatus and/or receiving apparatus of the present specification.
(39) In the example of FIG. **1**, various technical features described below may be performed. FIG. **1** relates to at least one station (STA). For example, STAs **110** and **120** of the present specification may also be called in various terms such as a mobile terminal, a wireless device, a wireless transmit/receive unit (WTRU), a user equipment (UE), a mobile station (MS), a mobile subscriber unit, or simply a user. The STAs **110** and **120** of the present specification may also be called in various terms such as a network, a base station, a node-B, an access point (AP), a repeater, a router, a relay, or the like. The STAs **110** and **120** of the present specification may also be referred to as various names such as a receiving apparatus, a transmitting apparatus, a receiving STA, a transmitting STA, a receiving device, a transmitting device, or the like.
(40) For example, the STAs **110** and **120** may serve as an AP or a non-AP. That is, the STAs **110** and **120** of the present specification may serve as the AP and/or the non-AP.
(41) The STAs **110** and **120** of the present specification may support various communication standards together in addition to the IEEE 802.11 standard. For example, a communication standard (e.g., LTE, LTE-A, 5G NR standard) or the like based on the 3GPP standard may be supported. In addition, the STA of the present specification may be implemented as various devices such as a mobile phone, a vehicle, a personal computer, or the like. In addition, the STA of the present specification may support communication for various communication services such as voice calls, video calls, data communication, and self-driving (autonomous-driving), or the like.
(42) The STAs **110** and **120** of the present specification may include a medium access control (MAC) conforming to the IEEE 802.11 standard and a physical layer interface for a radio medium.
(43) The STAs **110** and **120** will be described below with reference to a sub-figure (a) of FIG. **1**.
(44) The first STA **110** may include a processor **111**, a memory **112**, and a transceiver **113**. The illustrated process, memory, and transceiver may be implemented individually as separate chips, or at least two blocks/functions may be implemented through a single chip.
(45) The transceiver **113** of the first STA performs a signal transmission/reception operation. Specifically, an IEEE 802.11 packet (e.g., IEEE 802.11a/b/g/n/ac/ax/be, etc.) may be transmitted/received.
(46) For example, the first STA **110** may perform an operation intended by an AP. For example, the processor **111** of the AP may receive a signal through the transceiver **113**, process a reception (RX)

signal, generate a transmission (TX) signal, and provide control for signal transmission. The memory **112** of the AP may store a signal (e.g., RX signal) received through the transceiver **113**, and may store a signal (e.g., TX signal) to be transmitted through the transceiver.
(47) For example, the second STA **120** may perform an operation intended by a non-AP STA. For example, a transceiver **123** of a non-AP performs a signal transmission/reception operation. Specifically, an IEEE 802.11 packet (e.g., IEEE 802.11a/b/g/n/ac/ax/be packet, etc.) may be transmitted/received.
(48) For example, a processor **121** of the non-AP STA may receive a signal through the transceiver **123**, process an RX signal, generate a TX signal, and provide control for signal transmission. A memory **122** of the non-AP STA may store a signal (e.g., RX signal) received through the transceiver **123**, and may store a signal (e.g., TX signal) to be transmitted through the transceiver.
(49) For example, an operation of a device indicated as an AP in the specification described below may be performed in the first STA **110** or the second STA **120**. For example, if the first STA **110** is the AP, the operation of the device indicated as the AP may be controlled by the processor **111** of the first STA **110**, and a related signal may be transmitted or received through the transceiver **113** controlled by the processor **111** of the first STA **110**. In addition, control information related to the operation of the AP or a TX/RX signal of the AP may be stored in the memory **112** of the first STA **110**. In addition, if the second STA **120** is the AP, the operation of the device indicated as the AP may be controlled by the processor **121** of the second STA **120**, and a related signal may be transmitted or received through the transceiver **123** controlled by the processor **121** of the second STA **120**. In addition, control information related to the operation of the AP or a TX/RX signal of the AP may be stored in the memory **122** of the second STA **120**.
(50) For example, in the specification described below, an operation of a device indicated as a non-AP (or user-STA) may be performed in the first STA **110** or the second STA **120**. For example, if the second STA **120** is the non-AP, the operation of the device indicated as the non-AP may be controlled by the processor **121** of the second STA **120**, and a related signal may be transmitted or received through the transceiver **123** controlled by the processor **121** of the second STA **120**. In addition, control information related to the operation of the non-AP or a TX/RX signal of the non-AP may be stored in the memory **122** of the second STA **120**. For example, if the first STA **110** is the non-AP, the operation of the device indicated as the non-AP may be controlled by the processor **111** of the first STA **110**, and a related signal may be transmitted or received through the transceiver **113** controlled by the processor **111** of the first STA **110**. In addition, control information related to the operation of the non-AP or a TX/RX signal of the non-AP may be stored in the memory **112** of the first STA **110**.
(51) In the specification described below, a device called a (transmitting/receiving) STA, a first STA, a second STA, a STA1, a STA2, an AP, a first AP, a second AP, an AP1, an AP2, a (transmitting/receiving) terminal, a (transmitting/receiving) device, a (transmitting/receiving) apparatus, a network, or the like may imply the STAs **110** and **120** of FIG. **1**. For example, a device indicated as, without a specific reference numeral, the (transmitting/receiving) STA, the first STA, the second STA, the STA1, the STA2, the AP, the first AP, the second AP, the AP1, the AP2, the (transmitting/receiving) terminal, the (transmitting/receiving) device, the (transmitting/receiving) apparatus, the network, or the like may imply the STAs **110** and **120** of FIG. **1**. For example, in the following example, an operation in which various STAs transmit/receive a signal (e.g., a PPDU) may be performed in the transceivers **113** and **123** of FIG. **1**. In addition, in the following example, an operation in which various STAs generate a TX/RX signal or perform data processing and computation in advance for the TX/RX signal may be performed in the processors **111** and **121** of FIG. **1**. For example, an example of an operation for generating the TX/RX signal or performing the data processing and computation in advance may include: 1) an operation of determining/obtaining/configuring/computing/decoding/encoding bit information of a sub-field (SIG, STF, LTF, Data) included in a PPDU; 2) an operation of determining/configuring/obtaining a

time resource or frequency resource (e.g., a subcarrier resource) or the like used for the sub-field (SIG, STF, LTF, Data) included the PPDU; 3) an operation of determining/configuring/obtaining a specific sequence (e.g., a pilot sequence, an STF/LTF sequence, an extra sequence applied to SIG) or the like used for the sub-field (SIG, STF, LTF, Data) field included in the PPDU; 4) a power control operation and/or power saving operation applied for the STA; and 5) an operation related to determining/obtaining/configuring/decoding/encoding or the like of an ACK signal. In addition, in the following example, a variety of information used by various STAs for determining/obtaining/configuring/computing/decoding/decoding a TX/RX signal (e.g., information related to a field/subfield/control field/parameter/power or the like) may be stored in the memories **112** and **122** of FIG. **1**.

(52) The aforementioned device/STA of the sub-figure (a) of FIG. **1** may be modified as shown in the sub-figure (b) of FIG. **1**. Hereinafter, the STAs **110** and **120** of the present specification will be described based on the sub-figure (b) of FIG. **1**.

(53) For example, the transceivers **113** and **123** illustrated in the sub-figure (b) of FIG. **1** may perform the same function as the aforementioned transceiver illustrated in the sub-figure (a) of FIG. **1**. For example, processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1** may include the processors **111** and **121** and the memories **112** and **122**. The processors **111** and **121** and memories **112** and **122** illustrated in the sub-figure (b) of FIG. **1** may perform the same function as the aforementioned processors **111** and **121** and memories **112** and **122** illustrated in the sub-figure (a) of FIG. **1**.

(54) A mobile terminal, a wireless device, a wireless transmit/receive unit (WTRU), a user equipment (UE), a mobile station (MS), a mobile subscriber unit, a user, a user STA, a network, a base station, a Node-B, an access point (AP), a repeater, a router, a relay, a receiving unit, a transmitting unit, a receiving STA, a transmitting STA, a receiving device, a transmitting device, a receiving apparatus, and/or a transmitting apparatus, which are described below, may imply the STAs **110** and **120** illustrated in the sub-figure (a)/(b) of FIG. **1**, or may imply the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**. That is, a technical feature of the present specification may be performed in the STAs **110** and **120** illustrated in the sub-figure (a)/(b) of FIG. **1**, or may be performed only in the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**. For example, a technical feature in which the transmitting STA transmits a control signal may be understood as a technical feature in which a control signal generated in the processors **111** and **121** illustrated in the sub-figure (a)/(b) of FIG. **1** is transmitted through the transceivers **113** and **123** illustrated in the sub-figure (a)/(b) of FIG. **1**. Alternatively, the technical feature in which the transmitting STA transmits the control signal may be understood as a technical feature in which the control signal to be transferred to the transceivers **113** and **123** is generated in the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**.

(55) For example, a technical feature in which the receiving STA receives the control signal may be understood as a technical feature in which the control signal is received by means of the transceivers **113** and **123** illustrated in the sub-figure (a) of FIG. **1**. Alternatively, the technical feature in which the receiving STA receives the control signal may be understood as the technical feature in which the control signal received in the transceivers **113** and **123** illustrated in the sub-figure (a) of FIG. **1** is obtained by the processors **111** and **121** illustrated in the sub-figure (a) of FIG. **1**. Alternatively, the technical feature in which the receiving STA receives the control signal may be understood as the technical feature in which the control signal received in the transceivers **113** and **123** illustrated in the sub-figure (b) of FIG. **1** is obtained by the processing chips **114** and **124** illustrated in the sub-figure (b) of FIG. **1**.

(56) Referring to the sub-figure (b) of FIG. **1**, software codes **115** and **125** may be included in the memories **112** and **122**. The software codes **115** and **126** may include instructions for controlling an operation of the processors **111** and **121**. The software codes **115** and **125** may be included as various programming languages.

(57) The processors **111** and **121** or processing chips **114** and **124** of FIG. **1** may include an application-specific integrated circuit (ASIC), other chipsets, a logic circuit and/or a data processing device. The processor may be an application processor (AP). For example, the processors **111** and **121** or processing chips **114** and **124** of FIG. **1** may include at least one of a digital signal processor (DSP), a central processing unit (CPU), a graphics processing unit (GPU), and a modulator and demodulator (modem). For example, the processors **111** and **121** or processing chips **114** and **124** of FIG. **1** may be SNAPDRAGON™ series of processors made by Qualcomm®, EXYNOS™ series of processors made by Samsung®, A series of processors made by Apple®, HELIO™ series of processors made by MediaTek®, ATOM™ series of processors made by Intel® or processors enhanced from these processors.
(58) In the present specification, an uplink may imply a link for communication from a non-AP STA to an SP STA, and an uplink PPDU/packet/signal or the like may be transmitted through the uplink. In addition, in the present specification, a downlink may imply a link for communication from the AP STA to the non-AP STA, and a downlink PPDU/packet/signal or the like may be transmitted through the downlink.
(59) FIG. **2** is a conceptual view illustrating the structure of a wireless local area network (WLAN).
(60) An upper part of FIG. **2** illustrates the structure of an infrastructure basic service set (BSS) of institute of electrical and electronic engineers (IEEE) 802.11.
(61) Referring the upper part of FIG. **2**, the wireless LAN system may include one or more infrastructure BSSs **200** and **205** (hereinafter, referred to as BSS). The BSSs **200** and **205** as a set of an AP and a STA such as an access point (AP) **225** and a station (STA1) **200-1** which are successfully synchronized to communicate with each other are not concepts indicating a specific region. The BSS **205** may include one or more STAs **205-1** and **205-2** which may be joined to one AP **230**.
(62) The BSS may include at least one STA, APs providing a distribution service, and a distribution system (DS) **210** connecting multiple APs.
(63) The distribution system **210** may implement an extended service set (ESS) **240** extended by connecting the multiple BSSs **200** and **205**. The ESS **240** may be used as a term indicating one network configured by connecting one or more APs **225** or **230** through the distribution system **210**. The AP included in one ESS **240** may have the same service set identification (S SID).
(64) A portal **220** may serve as a bridge which connects the wireless LAN network (IEEE 802.11) and another network (e.g., 802.X).
(65) In the BSS illustrated in the upper part of FIG. **2**, a network between the APs **225** and **230** and a network between the APs **225** and **230** and the STAs **200-1**, **205-1**, and **205-2** may be implemented. However, the network is configured even between the STAs without the APs **225** and **230** to perform communication. A network in which the communication is performed by configuring the network even between the STAs without the APs **225** and **230** is defined as an Ad-Hoc network or an independent basic service set (IBSS).
(66) A lower part of FIG. **2** illustrates a conceptual view illustrating the IBSS.
(67) Referring to the lower part of FIG. **2**, the IBSS is a BSS that operates in an Ad-Hoc mode. Since the IBSS does not include the access point (AP), a centralized management entity that performs a management function at the center does not exist. That is, in the IBSS, STAs **250-1**, **250-2**, **250-3**, **255-4**, and **255-5** are managed by a distributed manner. In the IBSS, all STAs **250-1**, **250-2**, **250-3**, **255-4**, and **255-5** may be constituted by movable STAs and are not permitted to access the DS to constitute a self-contained network.
(68) FIG. **3** illustrates a general link setup process.
(69) In S**310**, a STA may perform a network discovery operation. The network discovery operation may include a scanning operation of the STA. That is, to access a network, the STA needs to discover a participating network. The STA needs to identify a compatible network before participating in a wireless network, and a process of identifying a network present in a particular

area is referred to as scanning. Scanning methods include active scanning and passive scanning.
(70) FIG. **3** illustrates a network discovery operation including an active scanning process. In active scanning, a STA performing scanning transmits a probe request frame and waits for a response to the probe request frame in order to identify which AP is present around while moving to channels. A responder transmits a probe response frame as a response to the probe request frame to the STA having transmitted the probe request frame. Here, the responder may be a STA that transmits the last beacon frame in a BSS of a channel being scanned. In the BSS, since an AP transmits a beacon frame, the AP is the responder. In an IBSS, since STAs in the IBSS transmit a beacon frame in turns, the responder is not fixed. For example, when the STA transmits a probe request frame via channel **1** and receives a probe response frame via channel **1**, the STA may store BSS-related information included in the received probe response frame, may move to the next channel (e.g., channel **2**), and may perform scanning (e.g., transmits a probe request and receives a probe response via channel **2**) by the same method.
(71) Although not shown in FIG. **3**, scanning may be performed by a passive scanning method. In passive scanning, a STA performing scanning may wait for a beacon frame while moving to channels. A beacon frame is one of management frames in IEEE 802.11 and is periodically transmitted to indicate the presence of a wireless network and to enable the STA performing scanning to find the wireless network and to participate in the wireless network. In a BSS, an AP serves to periodically transmit a beacon frame. In an IBSS, STAs in the IBSS transmit a beacon frame in turns. Upon receiving the beacon frame, the STA performing scanning stores information related to a BSS included in the beacon frame and records beacon frame information in each channel while moving to another channel. The STA having received the beacon frame may store BSS-related information included in the received beacon frame, may move to the next channel, and may perform scanning in the next channel by the same method.
(72) After discovering the network, the STA may perform an authentication process in S**320**. The authentication process may be referred to as a first authentication process to be clearly distinguished from the following security setup operation in S**340**. The authentication process in S**320** may include a process in which the STA transmits an authentication request frame to the AP and the AP transmits an authentication response frame to the STA in response. The authentication frames used for an authentication request/response are management frames.
(73) The authentication frames may include information related to an authentication algorithm number, an authentication transaction sequence number, a status code, a challenge text, a robust security network (RSN), and a finite cyclic group.
(74) The STA may transmit the authentication request frame to the AP. The AP may determine whether to allow the authentication of the STA based on the information included in the received authentication request frame. The AP may provide the authentication processing result to the STA via the authentication response frame.
(75) When the STA is successfully authenticated, the STA may perform an association process in S**330**. The association process includes a process in which the STA transmits an association request frame to the AP and the AP transmits an association response frame to the STA in response. The association request frame may include, for example, information related to various capabilities, a beacon listen interval, a service set identifier (SSID), a supported rate, a supported channel, RSN, a mobility domain, a supported operating class, a traffic indication map (TIM) broadcast request, and an interworking service capability. The association response frame may include, for example, information related to various capabilities, a status code, an association ID (AID), a supported rate, an enhanced distributed channel access (EDCA) parameter set, a received channel power indicator (RCPI), a received signal-to-noise indicator (RSNI), a mobility domain, a timeout interval (association comeback time), an overlapping BSS scanning parameter, a TIM broadcast response, and a QoS map.
(76) In S**340**, the STA may perform a security setup process. The security setup process in S**340**

may include a process of setting up a private key through four-way handshaking, for example, through an extensible authentication protocol over LAN (EAPOL) frame.
(77) FIG. **4** illustrates an example of a PPDU used in an IEEE standard.
(78) As illustrated, various types of PHY protocol data units (PPDUs) are used in IEEE a/g/n/ac standards. Specifically, an LTF and a STF include a training signal, a SIG-A and a SIG-B include control information for a receiving STA, and a data field includes user data corresponding to a PSDU (MAC PDU/aggregated MAC PDU).
(79) FIG. **4** also includes an example of an HE PPDU according to IEEE 802.11ax. The HE PPDU according to FIG. **4** is an illustrative PPDU for multiple users. An HE-SIG-B may be included only in a PPDU for multiple users, and an HE-SIG-B may be omitted in a PPDU for a single user.
(80) As illustrated in FIG. **4**, the HE-PPDU for multiple users (MUs) may include a legacy-short training field (L-STF), a legacy-long training field (L-LTF), a legacy-signal (L-SIG), a high efficiency-signal A (HE-SIG A), a high efficiency-signal-B (HE-SIG B), a high efficiency-short training field (HE-STF), a high efficiency-long training field (HE-LTF), a data field (alternatively, an MAC payload), and a packet extension (PE) field. The respective fields may be transmitted for illustrated time periods (i.e., 4 or 8 μs).
(81) Hereinafter, a resource unit (RU) used for a PPDU is described. An RU may include a plurality of subcarriers (or tones). An RU may be used to transmit a signal to a plurality of STAs according to OFDMA. Further, an RU may also be defined to transmit a signal to one STA. An RU may be used for an STF, an LTF, a data field, or the like.
(82) The RU described in the present specification may be used in uplink (UL) communication and downlink (DL) communication. For example, when UL-MU communication which is solicited by a trigger frame is performed, a transmitting STA (e.g., an AP) may allocate a first RU (e.g., 26/52/106/242-RU, etc.) to a first STA through the trigger frame, and may allocate a second RU (e.g., 26/52/106/242-RU, etc.) to a second STA. Thereafter, the first STA may transmit a first trigger-based PPDU based on the first RU, and the second STA may transmit a second trigger-based PPDU based on the second RU. The first/second trigger-based PPDU is transmitted to the AP at the same (or overlapped) time period.
(83) For example, when a DL MU PPDU is configured, the transmitting STA (e.g., AP) may allocate the first RU (e.g., 26/52/106/242-RU, etc.) to the first STA, and may allocate the second RU (e.g., 26/52/106/242-RU, etc.) to the second STA. That is, the transmitting STA (e.g., AP) may transmit HE-STF, HE-LTF, and Data fields for the first STA through the first RU in one MU PPDU, and may transmit HE-STF, HE-LTF, and Data fields for the second STA through the second RU.
(84) FIG. **5** illustrates an operation based on UL-MU. As illustrated, a transmitting STA (e.g., an AP) may perform channel access through contending (e.g., a backoff operation), and may transmit a trigger frame **1030**. That is, the transmitting STA may transmit a PPDU including the trigger frame **1030**. Upon receiving the PPDU including the trigger frame, a trigger-based (TB) PPDU is transmitted after a delay corresponding to SIFS.
(85) TB PPDUs **1041** and **1042** may be transmitted at the same time period, and may be transmitted from a plurality of STAs (e.g., user STAs) having AIDs indicated in the trigger frame **1030**. An ACK frame **1050** for the TB PPDU may be implemented in various forms.
(86) A specific feature of the trigger frame is described with reference to FIG. **6** to FIG. **8**. Even if UL-MU communication is used, an orthogonal frequency division multiple access (OFDMA) scheme or a MU MIMO scheme may be used, and the OFDMA and MU-MIMO schemes may be simultaneously used.
(87) FIG. **6** illustrates an example of a trigger frame. The trigger frame of FIG. **6** allocates a resource for uplink multiple-user (MU) transmission, and may be transmitted, for example, from an AP. The trigger frame may be configured of a MAC frame, and may be included in a PPDU.
(88) Each field shown in FIG. **6** may be partially omitted, and another field may be added. In addition, a length of each field may be changed to be different from that shown in the figure.

(89) A frame control field **1110** of FIG. **6** may include information related to a MAC protocol version and extra additional control information. A duration field **1120** may include time information for NAV configuration or information related to an identifier (e.g., AID) of a STA.
(90) In addition, an RA field **1130** may include address information of a receiving STA of a corresponding trigger frame, and may be optionally omitted. A TA field **1140** may include address information of a STA (e.g., an AP) which transmits the corresponding trigger frame. A common information field **1150** includes common control information applied to the receiving STA which receives the corresponding trigger frame. For example, a field indicating a length of an L-SIG field of an uplink PPDU transmitted in response to the corresponding trigger frame or information for controlling content of a SIG-A field (i.e., HE-SIG-A field) of the uplink PPDU transmitted in response to the corresponding trigger frame may be included. In addition, as common control information, information related to a length of a CP of the uplink PPDU transmitted in response to the corresponding trigger frame or information related to a length of an LTF field may be included.
(91) In addition, per user information fields **1160** #**1** to **1160** #N corresponding to the number of receiving STAs which receive the trigger frame of FIG. **6** are preferably included. The per user information field may also be called an “allocation field”.
(92) In addition, the trigger frame of FIG. **6** may include a padding field **1170** and a frame check sequence field **1180**.
(93) Each of the per user information fields **1160** #**1** to **1160** #N shown in FIG. **6** may include a plurality of subfields.
(94) FIG. **7** illustrates an example of a common information field of a trigger frame. A subfield of FIG. **7** may be partially omitted, and an extra subfield may be added. In addition, a length of each subfield illustrated may be changed.
(95) A length field **1210** illustrated has the same value as a length field of an L-SIG field of an uplink PPDU transmitted in response to a corresponding trigger frame, and a length field of the L-SIG field of the uplink PPDU indicates a length of the uplink PPDU. As a result, the length field **1210** of the trigger frame may be used to indicate the length of the corresponding uplink PPDU.
(96) In addition, a cascade identifier field **1220** indicates whether a cascade operation is performed. The cascade operation implies that downlink MU transmission and uplink MU transmission are performed together in the same TXOP. That is, it implies that downlink MU transmission is performed and thereafter uplink MU transmission is performed after a pre-set time (e.g., SIFS). During the cascade operation, only one transmitting device (e.g., AP) may perform downlink communication, and a plurality of transmitting devices (e.g., non-APs) may perform uplink communication.
(97) A CS request field **1230** indicates whether a wireless medium state or a NAV or the like is necessarily considered in a situation where a receiving device which has received a corresponding trigger frame transmits a corresponding uplink PPDU.
(98) An HE-SIG-A information field **1240** may include information for controlling content of a SIG-A field (i.e., HE-SIG-A field) of the uplink PPDU in response to the corresponding trigger frame.
(99) A CP and LTF type field **1250** may include information related to a CP length and LTF length of the uplink PPDU transmitted in response to the corresponding trigger frame. A trigger type field **1260** may indicate a purpose of using the corresponding trigger frame, for example, typical triggering, triggering for beamforming, a request for block ACK/NACK, or the like.
(100) It may be assumed that the trigger type field **1260** of the trigger frame in the present specification indicates a trigger frame of a basic type for typical triggering. For example, the trigger frame of the basic type may be referred to as a basic trigger frame.
(101) FIG. **8** illustrates an example of a subfield included in a per user information field. A user information field **1300** of FIG. **8** may be understood as any one of the per user information fields **1160** #**1** to **1160** #N mentioned above with reference to FIG. **6**. A subfield included in the user

information field **1300** of FIG. **8** may be partially omitted, and an extra subfield may be added. In addition, a length of each subfield illustrated may be changed.
(102) A user identifier field **1310** of FIG. **8** indicates an identifier of a STA (i.e., receiving STA) corresponding to per user information. An example of the identifier may be the entirety or part of an association identifier (AID) value of the receiving STA.
(103) In addition, an RU allocation field **1320** may be included. That is, when the receiving STA identified through the user identifier field **1310** transmits a TB PPDU in response to the trigger frame, the TB PPDU is transmitted through an RU indicated by the RU allocation field **1320**.
(104) The subfield of FIG. **8** may include a coding type field **1330**. The coding type field **1330** may indicate a coding type of the TB PPDU. For example, when BCC coding is applied to the TB PPDU, the coding type field **1330** may be set to '1', and when LDPC coding is applied, the coding type field **1330** may be set to '0'.
(105) In addition, the subfield of FIG. **8** may include an MCS field **1340**. The MCS field **1340** may indicate an MCS scheme applied to the TB PPDU. For example, when BCC coding is applied to the TB PPDU, the coding type field **1330** may be set to '1', and when LDPC coding is applied, the coding type field **1330** may be set to '0'.
(106) Hereinafter, a UL OFDMA-based random access (UORA) scheme will be described.
(107) FIG. **9** describes a technical feature of the UORA scheme.
(108) A transmitting STA (e.g., an AP) may allocate six RU resources through a trigger frame as shown in FIG. **9**. Specifically, the AP may allocate a 1st RU resource (AID 0, RU 1), a 2nd RU resource (AID 0, RU 2), a 3rd RU resource (AID 0, RU 3), a 4th RU resource (AID 2045, RU 4), a 5th RU resource (AID 2045, RU 5), and a 6th RU resource (AID 3, RU 6). Information related to the AID 0, AID 3, or AID 2045 may be included, for example, in the user identifier field **1310** of FIG. **8**. Information related to the RU 1 to RU 6 may be included, for example, in the RU allocation field **1320** of FIG. **8**. AID=0 may imply a UORA resource for an associated STA, and AID=2045 may imply a UORA resource for an un-associated STA. Accordingly, the 1st to 3rd RU resources of FIG. **9** may be used as a UORA resource for the associated STA, the 4th and 5th RU resources of FIG. **9** may be used as a UORA resource for the un-associated STA, and the 6th RU resource of FIG. **9** may be used as a typical resource for UL MU.
(109) In the example of FIG. **9**, an OFDMA random access backoff (OBO) of a STA1 is decreased to 0, and the STA1 randomly selects the 2nd RU resource (AID 0, RU 2). In addition, since an OBO counter of a STA2/3 is greater than 0, an uplink resource is not allocated to the STA2/3. In addition, regarding a STA4 in FIG. **9**, since an AID (e.g., AID=3) of the STA4 is included in a trigger frame, a resource of the RU 6 is allocated without backoff.
(110) Specifically, since the STA1 of FIG. **9** is an associated STA, the total number of eligible RA RUs for the STA1 is 3 (RU 1, RU 2, and RU 3), and thus the STA1 decreases an OBO counter by 3 so that the OBO counter becomes 0. In addition, since the STA2 of FIG. **9** is an associated STA, the total number of eligible RA RUs for the STA2 is 3 (RU 1, RU 2, and RU 3), and thus the STA2 decreases the OBO counter by 3 but the OBO counter is greater than 0. In addition, since the STA3 of FIG. **9** is an un-associated STA, the total number of eligible RA RUs for the STA3 is 2 (RU 4, RU 5), and thus the STA3 decreases the OBO counter by 2 but the OBO counter is greater than 0.
(111) Hereinafter, a PPDU transmitted/received in a STA of the present specification will be described.
(112) FIG. **10** illustrates an example of a PPDU used in the present specification.
(113) The PPDU of FIG. **10** may be called in various terms such as an EHT PPDU, a TX PPDU, an RX PPDU, a first type or N-th type PPDU, or the like. For example, in the present specification, the PPDU or the EHT PPDU may be called in various terms such as a TX PPDU, a RX PPDU, a first type or N-th type PPDU, or the like. In addition, the EHT PPDU may be used in an EHT system and/or a new WLAN system enhanced from the EHT system.
(114) The PPDU of FIG. **10** may indicate the entirety or part of a PPDU type used in the EHT

system. For example, the example of FIG. **10** may be used for both of a single-user (SU) mode and a multi-user (MU) mode. In other words, the PPDU of FIG. **10** may be a PPDU for one receiving STA or a plurality of receiving STAs. When the PPDU of FIG. **10** is used for a trigger-based (TB) mode, the EHT-SIG of FIG. **10** may be omitted. In other words, an STA which has received a trigger frame for uplink-MU (UL-MU) may transmit the PPDU in which the EHT-SIG is omitted in the example of FIG. **10**.

(115) In FIG. **10**, an L-STF to an EHT-LTF may be called a preamble or a physical preamble, and may be generated/transmitted/received/obtained/decoded in a physical layer.

(116) A subcarrier spacing of the L-STF, L-LTF, L-SIG, RL-SIG, U-SIG, and EHT-SIG fields of FIG. **10** may be determined as 312.5 kHz, and a subcarrier spacing of the EHT-STF, EHT-LTF, and Data fields may be determined as 78.125 kHz. That is, a tone index (or subcarrier index) of the L-STF, L-LTF, L-SIG, RL-SIG, U-SIG, and EHT-SIG fields may be expressed in unit of 312.5 kHz, and a tone index (or subcarrier index) of the EHT-STF, EHT-LTF, and Data fields may be expressed in unit of 78.125 kHz.

(117) In the PPDU of FIG. **10**, the L-LTE and the L-STF may be the same as those in the conventional fields.

(118) The L-SIG field of FIG. **10** may include, for example, bit information of 24 bits. For example, the 24-bit information may include a rate field of 4 bits, a reserved bit of 1 bit, a length field of 12 bits, a parity bit of 1 bit, and a tail bit of 6 bits. For example, the length field of 12 bits may include information related to a length or time duration of a PPDU. For example, the length field of 12 bits may be determined based on a type of the PPDU. For example, when the PPDU is a non-HT, HT, VHT PPDU or an EHT PPDU, a value of the length field may be determined as a multiple of 3. For example, when the PPDU is an HE PPDU, the value of the length field may be determined as “a multiple of 3”+1 or “a multiple of 3”+2. In other words, for the non-HT, HT, VHT PPDI or the EHT PPDU, the value of the length field may be determined as a multiple of 3, and for the HE PPDU, the value of the length field may be determined as “a multiple of 3”+1 or “a multiple of 3”+2.

(119) For example, the transmitting STA may apply BCC encoding based on a ½ coding rate to the 24-bit information of the L-SIG field. Thereafter, the transmitting STA may obtain a BCC coding bit of 48 bits. BPSK modulation may be applied to the 48-bit coding bit, thereby generating 48 BPSK symbols. The transmitting STA may map the 48 BPSK symbols to positions except for a pilot subcarrier{subcarrier index −21, −7, +7, +21} and a DC subcarrier{subcarrier index 0}. As a result, the 48 BPSK symbols may be mapped to subcarrier indices −26 to −22, −20 to −8, −6 to −1, +1 to +6, +8 to +20, and +22 to +26. The transmitting STA may additionally map a signal of {−1, −1, −1, 1} to a subcarrier index{−28, −27, +27, +28}. The aforementioned signal may be used for channel estimation on a frequency domain corresponding to {−28, −27, +27, +28}.

(120) The transmitting STA may generate an RL-SIG generated in the same manner as the L-SIG. BPSK modulation may be applied to the RL-SIG. The receiving STA may know that the RX PPDU is the HE PPDU or the EHT PPDU, based on the presence of the RL-SIG.

(121) A universal SIG (U-SIG) may be inserted after the RL-SIG of FIG. **10**. The U-SIB may be called in various terms such as a first SIG field, a first SIG, a first type SIG, a control signal, a control signal field, a first (type) control signal, or the like.

(122) The U-SIG may include information of N bits, and may include information for identifying a type of the EHT PPDU. For example, the U-SIG may be configured based on two symbols (e.g., two contiguous OFDM symbols). Each symbol (e.g., OFDM symbol) for the U-SIG may have a duration of 4 μs. Each symbol of the U-SIG may be used to transmit the 26-bit information. For example, each symbol of the U-SIG may be transmitted/received based on 52 data tomes and 4 pilot tones.

(123) Through the U-SIG (or U-SIG field), for example, A-bit information (e.g., 52 un-coded bits) may be transmitted. A first symbol of the U-SIG may transmit first X-bit information (e.g., 26 un-

coded bits) of the A-bit information, and a second symbol of the U-SIB may transmit the remaining Y-bit information (e.g. 26 un-coded bits) of the A-bit information. For example, the transmitting STA may obtain 26 un-coded bits included in each U-SIG symbol. The transmitting STA may perform convolutional encoding (i.e., BCC encoding) based on a rate of R=½ to generate 52-coded bits, and may perform interleaving on the 52-coded bits. The transmitting STA may perform BPSK modulation on the interleaved 52-coded bits to generate 52 BPSK symbols to be allocated to each U-SIG symbol. One U-SIG symbol may be transmitted based on 65 tones (subcarriers) from a subcarrier index −28 to a subcarrier index +28, except for a DC index 0. The 52 BPSK symbols generated by the transmitting STA may be transmitted based on the remaining tones (subcarriers) except for pilot tones, i.e., tones −21, −7, +7, +21.

(124) For example, the A-bit information (e.g., 52 un-coded bits) generated by the U-SIG may include a CRC field (e.g., a field having a length of 4 bits) and a tail field (e.g., a field having a length of 6 bits). The CRC field and the tail field may be transmitted through the second symbol of the U-SIG. The CRC field may be generated based on 26 bits allocated to the first symbol of the U-SIG and the remaining 16 bits except for the CRC/tail fields in the second symbol, and may be generated based on the conventional CRC calculation algorithm. In addition, the tail field may be used to terminate trellis of a convolutional decoder, and may be set to, for example, "000000".

(125) The A-bit information (e.g., 52 un-coded bits) transmitted by the U-SIG (or U-SIG field) may be divided into version-independent bits and version-dependent bits. For example, the version-independent bits may have a fixed or variable size. For example, the version-independent bits may be allocated only to the first symbol of the U-SIG, or the version-independent bits may be allocated to both of the first and second symbols of the U-SIG. For example, the version-independent bits and the version-dependent bits may be called in various terms such as a first control bit, a second control bit, or the like.

(126) For example, the version-independent bits of the U-SIG may include a PHY version identifier of 3 bits. For example, the PHY version identifier of 3 bits may include information related to a PHY version of a TX/RX PPDU. For example, a first value of the PHY version identifier of 3 bits may indicate that the TX/RX PPDU is an EHT PPDU. In other words, when the transmitting STA transmits the EHT PPDU, the PHY version identifier of 3 bits may be set to a first value. In other words, the receiving STA may determine that the RX PPDU is the EHT PPDU, based on the PHY version identifier having the first value.

(127) For example, the version-independent bits of the U-SIG may include a UL/DL flag field of 1 bit. A first value of the UL/DL flag field of 1 bit relates to UL communication, and a second value of the UL/DL flag field relates to DL communication.

(128) For example, the version-independent bits of the U-SIG may include information related to a TXOP length and information related to a BSS color ID.

(129) For example, when the EHT PPDU is divided into various types (e.g., various types such as an EHT PPDU related to an SU mode, an EHT PPDU related to a MU mode, an EHT PPDU related to a TB mode, an EHT PPDU related to extended range transmission, or the like), information related to the type of the EHT PPDU may be included in the version-dependent bits of the U-SIG.

(130) For example, the U-SIG may include: 1) a bandwidth field including information related to a bandwidth; 2) a field including information related to an MCS scheme applied to EHT-SIG; 3) an indication field including information regarding whether a dual subcarrier modulation (DCM) scheme is applied to EHT-SIG; 4) a field including information related to the number of symbol used for EHT-SIG; 5) a field including information regarding whether the EHT-SIG is generated across a full band; 6) a field including information related to a type of EHT-LTF/STF; and 7) information related to a field indicating an EHT-LTF length and a CP length.

(131) In the following example, a signal represented as a (TX/RX/UL/DL) signal, a (TX/RX/UL/DL) frame, a (TX/RX/UL/DL) packet, a (TX/RX/UL/DL) data unit, (TX/RX/UL/DL)

data, or the like may be a signal transmitted/received based on the PPDU of FIG. **10**. The PPDU of FIG. **10** may be used to transmit/receive frames of various types. For example, the PPDU of FIG. **10** may be used for a control frame. An example of the control frame may include a request to send (RTS), a clear to send (CTS), a power save-poll (PS-poll), BlockACKReq, BlockAck, a null data packet (NDP) announcement, and a trigger frame. For example, the PPDU of FIG. **10** may be used for a management frame. An example of the management frame may include a beacon frame, a (re-)association request frame, a (re-)association response frame, a probe request frame, and a probe response frame. For example, the PPDU of FIG. **10** may be used for a data frame. For example, the PPDU of FIG. **10** may be used to simultaneously transmit at least two or more of the control frames, the management frame, and the data frame.

(132) FIG. **11** illustrates an example of a modified transmission device and/or receiving device of the present specification.

(133) Each device/STA of the sub-figure (a)/(b) of FIG. **1** may be modified as shown in FIG. **11**. A transceiver **630** of FIG. **11** may be identical to the transceivers **113** and **123** of FIG. **1**. The transceiver **630** of FIG. **11** may include a receiver and a transmitter.

(134) A processor **610** of FIG. **11** may be identical to the processors **111** and **121** of FIG. **1**. Alternatively, the processor **610** of FIG. **11** may be identical to the processing chips **114** and **124** of FIG. **1**.

(135) A memory **620** of FIG. **11** may be identical to the memories **112** and **122** of FIG. **1**. Alternatively, the memory **620** of FIG. **11** may be a separate external memory different from the memories **112** and **122** of FIG. **1**.

(136) Referring to FIG. **11**, a power management module **611** manages power for the processor **610** and/or the transceiver **630**. A battery **612** supplies power to the power management module **611**. A display **613** outputs a result processed by the processor **610**. A keypad **614** receives inputs to be used by the processor **610**. The keypad **614** may be displayed on the display **613**. A SIM card **615** may be an integrated circuit which is used to securely store an international mobile subscriber identity (IMSI) and its related key, which are used to identify and authenticate subscribers on mobile telephony devices such as mobile phones and computers.

(137) Referring to FIG. **11**, a speaker **640** may output a result related to a sound processed by the processor **610**. A microphone **641** may receive an input related to a sound to be used by the processor **610**.

(138) Hereinafter, technical features of multi-link (ML) supported by the STA of the present specification will be described.

(139) STAs (AP and/or non-AP STA) of the present specification may support multi-link (ML) communication. ML communication may mean communication supporting a plurality of links. Links related to ML communication may include channels (e.g., 20/40/80/160/240/320 MHz channels) of the 2.4 GHz band, the 5 GHz band, and the 6 GHz band.

(140) A plurality of links used for ML communication may be set in various ways. For example, a plurality of links supported by one STA for ML communication may be a plurality of channels in the 2.4 GHz band, a plurality of channels in the 5 GHz band, and a plurality of channels in the 6 GHz band. Alternatively, a plurality of links may be a combination of at least one channel within the 2.4 GHz band (or 5 GHz/6 GHz band) and at least one channel within the 5 GHz band (or 2.4 GHz/6 GHz band). Meanwhile, at least one of a plurality of links supported by one STA for ML communication may be a channel to which preamble puncturing is applied.

(141) The STA may perform ML setup to perform ML communication. ML setup may be performed based on management frames or control frames such as Beacon, Probe Request/Response, and Association Request/Response. For example, information on ML setup may be included in element fields included in Beacon, Probe Request/Response, and Association Request/Response.

(142) When ML setup is completed, an enabled link for ML communication may be determined.

The STA may perform frame exchange through at least one of a plurality of links determined as an enabled link. For example, an enabled link may be used for at least one of a management frame, a control frame, and a data frame.
(143) When one STA supports a plurality of Links, a transmitting/receiving device supporting each Link may operate like one logical STA. For example, one STA supporting two links may be expressed as one ML device (Multi Link Device; MLD) including a first STA for a first link and a second STA for a second link. For example, one AP supporting two links may be expressed as one AP MLD including a first AP for a first link and a second AP for a second link. In addition, one non-AP supporting two links may be expressed as one non-AP MLD including a first STA for the first link and a second STA for the second link.
(144) More specific features of the ML setup are described below.
(145) An MLD (AP MLD and/or non-AP MLD) may transmit information about a link that the corresponding MLD can support through ML setup. Link-related information may be configured in various ways. For example, link-related information includes at least one of 1) information on whether the MLD (or STA) supports simultaneous RX/TX operation, 2) information on the number/upper limit of uplink/downlink links supported by the MLD (or STA), 3) information on the location/band/resource of uplink/downlink link supported by MLD (or STA), 4) type of frame available or preferred in at least one uplink/downlink link (management, control, data etc.), 5) available or preferred ACK policy information on at least one uplink/downlink link, and 6) information on available or preferred TID (traffic identifier) on at least one uplink/downlink link. The TID is related to the priority of traffic data and is represented by 8 types of values according to the conventional wireless LAN standard. That is, 8 TID values corresponding to 4 access categories (AC) (AC BK (background), AC BE (best effort), AC_VI (video), AC_VO (voice)) according to the conventional wireless LAN standard may be defined.
(146) For example, it may be set in advance that all TIDs are mapped for uplink/downlink links. Specifically, if negotiation is not done through ML setup, all TIDs may be used for ML communication, and if mapping between uplink/downlink links and TIDs is negotiated through additional ML setup, the negotiated TIDs may be used for ML communication.
(147) A plurality of links that can be used by the transmitting MLD and the receiving MLD related to ML communication can be set through ML setup, and this can be called an enabled link. The enabled link can be called differently in a variety of ways. For example, it may be called various expressions such as a first link, a second link, a transmitting link, and a receiving link.
(148) After the ML setup is complete, the MLD may update the ML setup. For example, the MLD may transmit information about a new link when updating information about a link is required. Information about the new link may be transmitted based on at least one of a management frame, a control frame, and a data frame.
(149) The device described below may be the apparatus of FIGS. **1** and/or **11**, and the PPDU may be the PPDU of FIG. **10**. A device may be an AP or a non-AP STA. A device described below may be an AP multi-link device (MLD) or a non-AP STA MLD supporting multi-link.
(150) In EHT (extremely high throughput), a standard being discussed after 802.11ax, a multi-link environment in which one or more bands are simultaneously used is considered. When a device supports multi-link, the device can simultaneously or alternately use one or more bands (e.g., 2.4 GHz, 5 GHz, 6 GHz, 60 GHz, etc.).
(151) In the following specification, MLD means a multi-link device. The MLD has one or more connected STAs and has one MAC service access point (SAP) that communicates with the upper link layer (Logical Link Control, LLC). MLD may mean a physical device or a logical device. Hereinafter, a device may mean an MLD.
(152) In the following specification, a transmitting device and a receiving device may mean MLD. The first link of the receiving/transmitting device may be a terminal (e.g., STA or AP) included in the receiving/transmitting device and performing signal transmission/reception through the first

link. The second link of the receiving/transmitting device may be a terminal (e.g., STA or AP) that transmits/receives a signal through the second link included in the receiving/transmitting device.
(153) In IEEE802.11be, two types of multi-link operations can be supported. For example, simultaneous transmit and receive (STR) and non-STR operations may be considered. For example, STR may be referred to as asynchronous multi-link operation, and non-STR may be referred to as synchronous multi-link operation. Multi-links may include multi-bands. That is, multi-links may mean links included in several frequency bands or may mean multiple links included in one frequency band.
(154) EHT (11be) considers multi-link technology, where multi-link may include multi-band. That is, multi-link can represent links of several bands and multiple multi-links within one band at the same time. Two major multi-link operations are being considered. Asynchronous operation, which enables TX/RX simultaneously on several links, and synchronous operation, which is not possible, are being considered. Hereinafter, a capability that enables simultaneous reception and transmission on multiple links is referred to as STR (simultaneous transmit and receive), an STA having STR capability is referred to as STR MLD (multi-link device), and an STA that does not have STR capability is referred to as a non-STR MLD.
(155) In the following specification, for convenience of explanation, it is described that the MLD (or the processor of the MLD) controls at least one STA, but is not limited thereto. As described above, the at least one STA may transmit and receive signals independently regardless of MLD.
(156) According to an embodiment, an AP MLD or a non-AP MLD may have a structure having a plurality of links. In other words, a non-AP MLD can support multiple links. A non-AP MLD may include a plurality of STAs. A plurality of STAs may have Link for each STA.
(157) In the EHT standard (802.11be standard), the MLD (Multi-Link Device) structure in which one AP/non-AP MLD supports multiple links is considered as a major technology. STAs included in the non-AP MLD may transmit information about other STAs in the non-AP MLD together through one link. Accordingly, there is an effect of reducing the overhead of frame exchange. In addition, there is an effect of increasing the link use efficiency of the STA and reducing power consumption.
(158) Here, multi-link may include multi-band. That is, multi-link can represent links of several bands and multiple multi-links within one band at the same time.
(159) FIG. **12** shows an example of a structure of a STA MLD.
(160) FIG. **12** shows an example in which one STA MLD has three links. In 802.11be, one STA in the STA MLD must provide information on one or more links it has for single setup.
(161) FIG. **13** shows an example of setting Multi-Link.
(162) FIG. **13** shows an example of Setup. AP MLD has 4 links, and information about them is announced through Beacon frame or Probe Response frame. The information may provide information on capabilities of AP/Links, channel information, and non-STR link set. This example shows that Links 1 and 2 and Links 3 and 4 are non-STR link sets, respectively. Non-AP MLD has 3 STAs, each STA can be connected through a link, and discovers the AP MLD through link 4. The non-AP MLD requests Link 2, 3, 4 through link 4 as the link to operate, and the AP MLD responds based on this request so that the non-AP MLD can operate on links 2, 3, 4. If, in this example, the STA MLD does not have the capability to operate on link 4, it may not request link 4.
(163) In the process of FIG. **13**, when three link pairs are {STA 1<->AP 4}, {STA 2<->AP 3}, and {STA 3<->AP 2} in Multi-link Setup, STA 2 and STA 3 have not yet performed an initial frame exchange with the AP. For example, when AP 2 receives a frame from STA 3, it does not know whether STA 3 belongs to the same MLD as STA 1. Therefore, AP 2 and AP 3 belong to the same MLD as STA 2 and STA 3, respectively, and a method for enabling frame exchange on the corresponding link is required.
(164) The methods mentioned above are as follows, but are not limited to one method.
(165) First of all, it can be largely divided into 1) method at the multi-link setup stage and 2)

method after multi-link setup. In these methods, among the links to be setup, the link transmitted in the Association Request/Response frame is referred to as an association link, and the other setup links are referred to as non-association links. For example, if the setup links in FIG. **13** are links 2, 3, and 4, link 4 becomes an association link and links 2 and 3 become non-association links.
(166) 1) Method at the Multi-Link Setup Stage
(167) 1-1) MAC Address Signaling: An STA transmitting a Management frame (i.e., Probe Request/Response frame, Beacon frame, Association Request/Response frame) includes MAC addresses of other STAs belonging to the same MLD other than itself. STAs operating on a non-association link recognize this MAC address to enable frame exchange. A method of including the MAC addresses of other STAs belonging to the same MLD is as follows, but is not limited thereto.
(168) A. New element or field definition: A new element or field is defined and MAC addresses of other STAs in the same MLD are included in the management frame. Basically, the MLD Per-STA MAC address field can be defined as shown in FIG. **14**.
(169) FIG. **14** shows an example of an MLD Per-STA MAC address field.
(170) In the case of AP MLD, since there is a link ID that can distinguish each AP, the MAC address can be indicated in the order of Link ID. Even in the case of a non-AP MLD, if an STA ID capable of distinguishing each STA is defined, the MAC address can be indicated in the order of the STAID. Also, for clear indication, link ID or STA ID may be indicated before each MAC address. In addition, MAC addresses for all STAs belonging to the same MLD other than the STA transmitting the management frame may not be indicated. For example, in FIG. **13**, STA 1 may not include the MAC address of STA 2. In this case, Number of STAs (or Number of links (in case of AP)) may be additionally indicated.
(171) If it is not an MLD, since the field may not be included, MAC addresses of other STAs may be included in the form of an element including the MLD address field shown in FIG. **14** instead of the field (see FIG. **15**).
(172) FIG. **15** shows an example of an MLD Per-STA MAC address element.
(173) The MAC Address subelement exists in the current baseline. Therefore, if it is entered in the form of a subelement, the existing MAC Address subelement can be utilized. Therefore, the MAC address element mentioned below may be replaced with a MAC address subelement.
(174) B. Inclusion in Multi-link element: MAC addresses of other STAs in the same MLD may be included in common info or per-STA info of the ML IE.
(175) The format of the Multi-link Element is as shown in FIGS. **16** and **17**, and the order, name, and size of the fields may change, and may exist as an additional field. Basically, Common info means common information between STAs in the MLD, and specific information on each STA is indicated in the Per-STA Profile.
(176) FIG. **16** illustrates structures of a Multi-link Control field and a Common Info field of a Multi-link Element.
(177) Referring to FIG. **16**, the Multi-link Control field includes a Type subfield and an MLD MAC Address Present subfield. The Common Info field includes the MLD MAC Address subfield. When the MLD MAC Address Present subfield is set to 1 (or 0), MAC addresses of STAs in the MLD may be included in the MLD MAC Address subfield.
(178) FIG. **17** shows the structure of the Link Info field of Multi-link Element.
(179) Referring to FIG. **17**, the Link Info field includes a Per-STA Profile subfield when the optional subelement ID is 0, and includes a Vendor Specific subfield when the optional subelement ID is 221. Optional subelement IDs for Multi-link Element are defined as follows.
(180) TABLE-US-00001 TABLE 1 Subelement ID Name Extensible   0 Per-STA Profile Yes   1-220 Reserved 221 Vendor Specific Vendor defined 222-255 Reserved
(181) The Link Info field includes a Per-STA Profile subfield for other STAs (STAs operating in the non-association link) within the same MLD. Referring to FIG. **17**, assuming that the STA MLD includes STA 2 and STA 3, the Link Info field may include a Per-STA Profile #**2** subfield for STA 2

and a Per-STA Profile #**3** subfield for STA 3.
(182) FIG. **18** shows an example in which the MLD Per-STA MAC address field is included in the Common Info field of the Multi-link Element.
(183) FIG. **19** shows an example in which the MAC address of the STA is included in the Per-STA Profile subfield of the Link Info field of the Multi-link Element.
(184) The MLD Per-STA MAC address field of FIG. **14** may be included in Common Info as shown in FIG. **18** or may be present as a MAC address field or element of each STA in Per-STA Info as shown in FIG. **19**. The Per-STA Profile may not be required if it exists in Common Info and the MAC address must always be included in the ML IE. However, since each MAC address is different for each STA, a Per-STA Profile may be semantically appropriate.
(185) FIG. **20** shows an example in which the Complete Profile is included in the Per-STA Profile subfield of the Link Info field of the Multi-link Element.
(186) When the MAC address of another STA in the same MLD is included in the Per-STA Profile, when Complete Profile=1 by default, the Per-STA Profile may include a complete profile as shown in FIG. **20**. In addition, the Complete Profile has several fields or several elements with a fixed order. Therefore, the MAC address included in the Per-STA Profile may be signaled in the form of one field or element among several fields or elements configured in a fixed order. For example, if it is in the form of an element, as shown in FIG. **20**, element **2** can be replaced with a MAC address (sub) element to include the MAC address of STA x.
(187) FIG. **21** shows an example of including MAC address present in the Per-STA Profile subfield of the Link Info field of the multi-link element.
(188) In addition, when several fields in a fixed order are included as one field, if included only in the case of association and not included in other cases to reduce overhead, signaling can be performed by including the presence field for the MAC address in the Per-STA Control field as shown in FIG. **21**. That is, if the value of the MAC address Present field is 1, the MAC address of the STA in the MLD corresponding to the Per-STA field exists.
(189) 2) Method after Multi-Link Setup
(190) 2-1) Initial Frame Transmission on Non-Association Link
(191) FIG. **22** shows an example of transmitting an Initial frame on a non-association link.
(192) Referring to FIG. **22**, to trigger frame exchange, the transmitter transmits an initial frame on a non-association link. When the Transmitter transmits the Initial Request frame, the receiver can respond through the Initial Response frame (e.g., ACK). A transmitter can be both an AP and a non-AP STA. For example, after multi-link setup, an STA that first obtains a channel access opportunity on a non-association link may transmit an Initial Request frame. Basically, when the initial frame is transmitted, the MAC address of the transmitter STA can be known through the TA.
(193) Initial Request frame can be defined as a new frame, but existing QoS Data frame, QoS Null frame, etc. can be used. In particular, the frame transmitted as the Initial Request frame may include transmitter MLD information and/or receiver MLD information in order to notify that operation is possible in the corresponding link after multi-link setup.
(194) A. Address Setting: MLD MAC address can be set in Initial frame.
(195) A-1) Set the MLD MAC address for the Receiver in the Receiving Address (RA) field:
(196) FIG. **23** shows another example of transmitting an Initial frame on a non-association link.
(197) Basically, in the multi-link setup process, since each other's MLD MAC address can be known by including the ML IE in the Association Request/Response frame, by including the MLD MAC address of the MLD, which is the receiver, in the RA, it can be notified that it is operational in the current link. In addition, in order to inform that it is an MLD that has performed multi-link setup, the MLD MAC address of the transmitter MLD may be included in the source address (SA). For example, as shown in FIG. **23**, when STA 2 transmits an initial request frame on link 3, TA can set its own MAC address, RA can set AP MLD MAC address, and SA can set MAC address of non-AP MLD. The example of A-1) follows the basic baseline, but there is an overhead that requires

additional use of the SA field.
(198) A-2) Broadcast address set in RA: From the receiver's point of view, the address can be known through the TA, but it is difficult to accurately distinguish whether it is an MLD with multi-link setup.
(199) A-3) Broadcast Address is Set in RA, and MLD MAC Address for Receiver is Set in Destination Address (DA):
(200) FIG. **24** shows another example of transmitting an Initial frame on a non-association link.
(201) It has the same purpose as A-1), but the address setting method is different. In addition, in order to inform that it is an MLD that has performed multi-link setup, the MLD MAC address of the transmitter MLD may be included in the source address (SA). For example, as shown in FIG. **24**, when STA 2 transmits an initial Request frame on link 3, TA can be configured as its own MAC address, RA can be configured as the broadcast address, SA can be configured as the MAC address of the non-AP MLD, and DA can be configured as the AP MLD MAC address. The example of A-3) follows the basic baseline, but there is overhead that requires additional use of SA and/or DA fields.
(202) A-4) Set the MLD MAC address for the transmitter of the transmitter MLD in the TA:
(203) FIG. **25** shows another example of transmitting an Initial frame on a non-association link.
(204) Although the purpose is the same as above, by setting the transmitter MLD address in the TA, the receiver who knows the MLD address can know to which MLD the corresponding transmitter STA belongs. In particular, this method can reduce overhead for the address field compared to the method of using SA and/or DA in A-1) to A-3). In the case of A-4), as shown in FIG. **25**, the RA basically uses the MAC address of the peer AP, but to clarify the intent of the initial frame, the MAC address of the MLD to which the AP belongs −1) or Broadcast address (A-2/A-3) can also be used.
(205) A-5) In addition, the transmitter transmits the initial frame including the MLD MAC address of the transmitter MLD and/or the receiver MLD in the control field (e.g. A-control field of QoS Null/Data frame) of the MAC frame body or MAC header delivered in the initial frame.
(206) B. MLD ID setting: MLD ID can be set in Initial frame.
(207) FIG. **26** shows another example of transmitting an Initial frame on a non-association link.
(208) B-1) The MLD MAC addresses of the transmitter and receiver set in the address fields presented in A-1), A-2), A-3), and A-4) may be replaced with MLD IDs that can differentiate MLD.
(209) For example, in the case of substitution in method A-4), the transmitter MLD and/or Transmits an initial frame including the MLD ID of the receiver MLD. In the case of MLD ID entering the address field, the size of the field is much smaller than that of the MLD MAC address, so the number of remaining bits may increase if entered in the corresponding field. In FIG. **26**, when STA 2 transmits an Initial Request frame on link 3, the transmitter puts the ID of the non-AP MLD in the A-Control field of this frame to indicate that it is the non-AP MLD of STA 2, and the receiver includes the ID of the corresponding AP MLD to indicate that it is the AP MLD of AP 4 (combination of method A-5 and method B-1)).
(210) C. Association ID setting: The non-AP STA of the Non-AP MLD may set the association ID in the Initial frame.
(211) FIG. **27** shows another example of transmitting an Initial frame on a non-association link.
(212) Specifically, as shown in A-4), the initial frame is transmitted including the Association Identifier (AID) of the non-AP STA of the non-AP MLD that transmits the initial frame in the control field (e.g. A-control field of QoS Null/Data frame) of the MAC frame body or MAC header delivered to the initial frame. If the AID is MLD-level, when the AID is included, it is possible to know the MLD to which the non-AP STA belongs and at the same time know the MAC address through the TA, the AP and the non-AP STA may exchange frames on a non-association link. In FIG. **27**, when STA 2 transmits an Initial Request frame on link 3, the transmitter puts the ID of the non-AP MLD in the A-Control field of this frame to indicate that it is the non-AP MLD of STA 2,

and the receiver includes the ID of the corresponding AP MLD to indicate that it is the AP MLD of AP 4.
(213) 2-2) Defining additional rules: Anon-AP STA does not transmit a frame until a beacon is received on a non-association link. That is, the non-AP STA waits until it receives (or hears) the Beacon. Basically, since TBTT (Target Beacon Transmission Time) information of other APs can be known during the multi-link setup process, this rule can be applied. For example, in FIG. **27**, STA 2 and STA 3 do not transmit a frame until receiving a beacon from AP 3 and AP 2, respectively. However, even if STA 2 transmits a frame to AP 3 first, AP 3 cannot know whether STA 2 belongs to the same MLD as STAs **1** and **3**.
(214) Hereinafter, the above-described embodiment will be described with reference to FIG. **1** to FIG. **27**.
(215) FIG. **28** is a flowchart illustrating a procedure in which a transmitting MLD provides MAC addresses of other STAs to a receiving MLD through a Multi-Link element (ML element) according to the present embodiment.
(216) The example of FIG. **28** may be performed in a network environment in which a next generation WLAN system (IEEE 802.11be or EHT WLAN system) is supported. The next generation wireless LAN system is a WLAN system that is enhanced from an 802.11ax system and may, therefore, satisfy backward compatibility with the 802.11ax system.
(217) The present embodiment proposes a method and apparatus for receiving, by a non-AP STA, a MAC address of another non-AP STA in the same non-AP MLD through an ML element in MLD communication. The non-AP STA recognizes another non-AP STA based on the MAC address of the other non-AP STA, so that frame exchange is possible on a corresponding link. Here, the transmitting MLD may correspond to an AP MLD, and the receiving MLD may correspond to a non-AP MLD. If the non-AP STA is a first receiving STA, a first transmitting STA connected to the first receiving STA through a first link may be referred to as a peer AP, the second to third transmitting STAs connected through different links may be referred to as other APs, and the second and third receiving STAs connected through different links may be referred to as other non-AP STAs.
(218) In step S**2810**, a transmitting multi-link device (MLD) generates a multi-link (ML) element.
(219) In step S**2820**, the transmitting MLD transmits the ML element to the receiving MLD through the first link.
(220) The transmitting MLD includes a first transmitting station (STA) operating on the first link and a second transmitting STA operating on a second link. The receiving MLD may include a first receiving STA operating on the first link and a second receiving STA operating on the second link.
(221) The ML element includes common information and link information.
(222) The link information includes a profile field of the second transmitting STA. The profile field of the second transmitting STA includes first information on whether a MAC address of the second transmitting STA is present. If the first information is set to 1, the profile field of the second transmitting STA includes the MAC address of the second transmitting STA. When the first information is set to 0, the profile field of the second transmitting STA may not include the MAC address of the second transmitting STA.
(223) The transmitting MLD may further include a third transmitting STA operating in a third link. The receiving MLD may further include a third transmitting STA operating in the third link.
(224) The link information may further include a profile field of the third transmitting STA. The profile field of the third transmitting STA may include second information on whether a MAC address of the third transmitting STA is present. When the second information is set to 1, the profile field of the third transmitting STA may include the MAC address of the third transmitting STA. When the second information is set to 0, the profile field of the third transmitting STA may not include the MAC address of the third transmitting STA.
(225) The first link may be an association link, and the second and third links may be non-

association links. That is, the present embodiment has an effect of enabling frame exchange in the non-association link by receiving the MAC address of the STA operating in the non-association link through the association link.
(226) The ML configuration process between the transmitting MLD and the receiving MLD is as follows.
(227) The transmitting MLD may transmit link setting information to the receiving MLD through the first link. The transmitting MLD may receive an association request frame from the receiving MLD through the first link. The transmitting MLD may transmit an association response frame to the transmitting MLD through the first link.
(228) The link setting information may be included in a beacon frame or a probe response frame. The link setting information may include link capabilities of the transmitting MLD, channel information, or information on a link set that cannot be simultaneously transmitted and received. The ML element may be included in the association response frame.
(229) A frame exchange process after ML configuration between the transmitting MLD and the receiving MLD is as follows.
(230) The transmitting MLD may receive a first initial request frame from the receiving MLD through the second link. The transmitting MLD may transmit a first initial response frame to the receiving MLD through the second link. The transmitting MLD may receive a second initiation request frame from the receiving MLD through the third link. The transmitting MLD may transmit a second initiation response frame to the receiving MLD through the third link.
(231) The first initiation request frame and the first initiation response frame may be exchanged in the second link based on the MAC address of the second transmitting STA. The second initiation request frame and the second initiation response frame may be exchanged in the third link based on a MAC address of the third transmitting STA.
(232) Also, the first and second initiation request frames may include MAC addresses of the transmitting and receiving MLDs, identifiers or association Identifiers (AIDs) of the transmitting and receiving MLDs.
(233) That is, this embodiment proposes a method in which the transmitting MLD delivers the MAC address of another transmitting STA in the receiving MLD through the profile field of the receiving STA included in the link information of the ML element. As a result, the first receiving STA recognizes the second and third receiving STAs in the ML configuration process. After ML configuration, frame exchange between the second receiving STA and the second transmitting STA and frame exchange between the third receiving STA and the third transmitting STA are enabled, and effective MLD communication can be performed.
(234) The ML element may further include an element identifier field, a length field, an element identifier extension field, and a Multi-Link (ML) control field.
(235) The ML control field may include a type field. If the type field is set to 0, the ML element may be a Basic ML element. If the type field is set to 1, the ML element may be a probe request ML element.
(236) FIG. **29** is a flowchart illustrating a procedure in which a receiving MLD receives MAC addresses of other STAs from a transmitting MLD through a Multi-Link element (ML element) according to the present embodiment.
(237) The example of FIG. **29** may be performed in a network environment in which a next generation WLAN system (IEEE 802.11be or EHT WLAN system) is supported. The next generation wireless LAN system is a WLAN system that is enhanced from an 802.11ax system and may, therefore, satisfy backward compatibility with the 802.11ax system.
(238) The present embodiment proposes a method and apparatus for requesting changed update information of a specific link by including a change sequence element in a multi-link element in a probe request frame in MLD communication.
(239) In step S**2910**, a receiving Multi-link Device (MLD) receives a multi-link (ML) element from

a transmitting MLD through a first link.
(240) In step S**2920**, the receiving MLD decodes the ML element.
(241) The transmitting MLD includes a first transmitting station (STA) operating on the first link and a second transmitting STA operating on a second link. The receiving MLD may include a first receiving STA operating on the first link and a second receiving STA operating on the second link.
(242) The ML element includes common information and link information.
(243) The link information includes a profile field of the second transmitting STA. The profile field of the second transmitting STA includes first information on whether a MAC address of the second transmitting STA is present. If the first information is set to 1, the profile field of the second transmitting STA includes the MAC address of the second transmitting STA. When the first information is set to 0, the profile field of the second transmitting STA may not include the MAC address of the second transmitting STA.
(244) The transmitting MLD may further include a third transmitting STA operating in a third link. The receiving MLD may further include a third transmitting STA operating in the third link.
(245) The link information may further include a profile field of the third transmitting STA. The profile field of the third transmitting STA may include second information on whether a MAC address of the third transmitting STA is present. When the second information is set to 1, the profile field of the third transmitting STA may include the MAC address of the third transmitting STA. When the second information is set to 0, the profile field of the third transmitting STA may not include the MAC address of the third transmitting STA.
(246) The first link may be an association link, and the second and third links may be non-association links. That is, the present embodiment has an effect of enabling frame exchange in the non-association link by receiving the MAC address of the STA operating in the non-association link through the association link.
(247) The ML configuration process between the transmitting MLD and the receiving MLD is as follows.
(248) The transmitting MLD may transmit link setting information to the receiving MLD through the first link. The transmitting MLD may receive an association request frame from the receiving MLD through the first link. The transmitting MLD may transmit an association response frame to the transmitting MLD through the first link.
(249) The link setting information may be included in a beacon frame or a probe response frame. The link setting information may include link capabilities of the transmitting MLD, channel information, or information on a link set that cannot be simultaneously transmitted and received. The ML element may be included in the association response frame.
(250) A frame exchange process after ML configuration between the transmitting MLD and the receiving MLD is as follows.
(251) The transmitting MLD may receive a first initial request frame from the receiving MLD through the second link. The transmitting MLD may transmit a first initial response frame to the receiving MLD through the second link. The transmitting MLD may receive a second initiation request frame from the receiving MLD through the third link. The transmitting MLD may transmit a second initiation response frame to the receiving MLD through the third link.
(252) The first initiation request frame and the first initiation response frame may be exchanged in the second link based on the MAC address of the second transmitting STA. The second initiation request frame and the second initiation response frame may be exchanged in the third link based on a MAC address of the third transmitting STA.
(253) Also, the first and second initiation request frames may include MAC addresses of the transmitting and receiving MLDs, identifiers or association Identifiers (AIDs) of the transmitting and receiving MLDs.
(254) That is, this embodiment proposes a method in which the transmitting MLD delivers the MAC address of another transmitting STA in the receiving MLD through the profile field of the

receiving STA included in the link information of the ML element. As a result, the first receiving STA recognizes the second and third receiving STAs in the ML configuration process. After ML configuration, frame exchange between the second receiving STA and the second transmitting STA and frame exchange between the third receiving STA and the third transmitting STA are enabled, and effective MLD communication can be performed.
(255) The ML element may further include an element identifier field, a length field, an element identifier extension field, and a Multi-Link (ML) control field.
(256) The ML control field may include a type field. If the type field is set to 0, the ML element may be a Basic ML element. If the type field is set to 1, the ML element may be a probe request ML element.
(257) The technical features of the present disclosure may be applied to various devices and methods. For example, the technical features of the present disclosure may be performed/supported through the device(s) of FIG. **1** and/or FIG. **11**. For example, the technical features of the present disclosure may be applied to only part of FIG. **1** and/or FIG. **11**. For example, the technical features of the present disclosure may be implemented based on the processing chip(s) **114** and **124** of FIG. **1**, or implemented based on the processor(s) **111** and **121** and the memory(s) **112** and **122**, or implemented based on the processor **610** and the memory **620** of FIG. **11**. For example, the device according to the present disclosure receives a multi-link (ML) element from a transmitting multi-link device (MLD) through a first link; and decodes the ML element.
(258) The technical features of the present disclosure may be implemented based on a computer readable medium (CRM). For example, a CRM according to the present disclosure is at least one computer readable medium including instructions designed to be executed by at least one processor.
(259) The CRM may store instructions that perform operations including receiving a multi-link (ML) element from a transmitting multi-link device (MLD) through a first link; and decoding the ML element. At least one processor may execute the instructions stored in the CRM according to the present disclosure. At least one processor related to the CRM of the present disclosure may be the processor **111**, **121** of FIG. **1**, the processing chip **114**, **124** of FIG. **1**, or the processor **610** of FIG. **11**. Meanwhile, the CRM of the present disclosure may be the memory **112**, **122** of FIG. **1**, the memory **620** of FIG. **11**, or a separate external memory/storage medium/disk.
(260) The foregoing technical features of the present specification are applicable to various applications or business models. For example, the foregoing technical features may be applied for wireless communication of a device supporting artificial intelligence (AI).
(261) Artificial intelligence refers to a field of study on artificial intelligence or methodologies for creating artificial intelligence, and machine learning refers to a field of study on methodologies for defining and solving various issues in the area of artificial intelligence. Machine learning is also defined as an algorithm for improving the performance of an operation through steady experiences of the operation.
(262) An artificial neural network (ANN) is a model used in machine learning and may refer to an overall problem-solving model that includes artificial neurons (nodes) forming a network by combining synapses. The artificial neural network may be defined by a pattern of connection between neurons of different layers, a learning process of updating a model parameter, and an activation function generating an output value.
(263) The artificial neural network may include an input layer, an output layer, and optionally one or more hidden layers. Each layer includes one or more neurons, and the artificial neural network may include synapses that connect neurons. In the artificial neural network, each neuron may output a function value of an activation function of input signals input through a synapse, weights, and deviations.
(264) A model parameter refers to a parameter determined through learning and includes a weight of synapse connection and a deviation of a neuron. A hyper-parameter refers to a parameter to be

set before learning in a machine learning algorithm and includes a learning rate, the number of iterations, a mini-batch size, and an initialization function.

(265) Learning an artificial neural network may be intended to determine a model parameter for minimizing a loss function. The loss function may be used as an index for determining an optimal model parameter in a process of learning the artificial neural network.

(266) Machine learning may be classified into supervised learning, unsupervised learning, and reinforcement learning.

(267) Supervised learning refers to a method of training an artificial neural network with a label given for training data, wherein the label may indicate a correct answer (or result value) that the artificial neural network needs to infer when the training data is input to the artificial neural network. Unsupervised learning may refer to a method of training an artificial neural network without a label given for training data. Reinforcement learning may refer to a training method for training an agent defined in an environment to choose an action or a sequence of actions to maximize a cumulative reward in each state.

(268) Machine learning implemented with a deep neural network (DNN) including a plurality of hidden layers among artificial neural networks is referred to as deep learning, and deep learning is part of machine learning. Hereinafter, machine learning is construed as including deep learning.

(269) The foregoing technical features may be applied to wireless communication of a robot.

(270) Robots may refer to machinery that automatically process or operate a given task with own ability thereof. In particular, a robot having a function of recognizing an environment and autonomously making a judgment to perform an operation may be referred to as an intelligent robot.

(271) Robots may be classified into industrial, medical, household, military robots and the like according uses or fields. A robot may include an actuator or a driver including a motor to perform various physical operations, such as moving a robot joint. In addition, a movable robot may include a wheel, a brake, a propeller, and the like in a driver to run on the ground or fly in the air through the driver.

(272) The foregoing technical features may be applied to a device supporting extended reality.

(273) Extended reality collectively refers to virtual reality (VR), augmented reality (AR), and mixed reality (MR). VR technology is a computer graphic technology of providing a real-world object and background only in a CG image, AR technology is a computer graphic technology of providing a virtual CG image on a real object image, and MR technology is a computer graphic technology of providing virtual objects mixed and combined with the real world.

(274) MR technology is similar to AR technology in that a real object and a virtual object are displayed together. However, a virtual object is used as a supplement to a real object in AR technology, whereas a virtual object and a real object are used as equal statuses in MR technology.

(275) XR technology may be applied to a head-mount display (HMD), a head-up display (HUD), a mobile phone, a tablet PC, a laptop computer, a desktop computer, a TV, digital signage, and the like. A device to which XR technology is applied may be referred to as an XR device.

(276) The claims recited in the present specification may be combined in a variety of ways. For example, the technical features of the method claims of the present specification may be combined to be implemented as a device, and the technical features of the device claims of the present specification may be combined to be implemented by a method. In addition, the technical characteristics of the method claim of the present specification and the technical characteristics of the device claim may be combined to be implemented as a device, and the technical characteristics of the method claim of the present specification and the technical characteristics of the device claim may be combined to be implemented by a method.

Claims

1. A method performed by an access point (AP) multi-link device (MLD) in a wireless local area network (WLAN) system, the method comprising: receiving, by a first AP affiliated with the AP MLD, a multi-link (ML) element from a first non-AP station (STA) affiliated with a non-AP MLD; and decoding, by the first AP, the ML element, wherein the first AP operating on a first link and a second AP operating on a second link are affiliated with the AP MLD, wherein the first non-AP STA operating on the first link and a second non-AP STA operating on the second link are affiliated with the non-AP MLD, wherein the ML element includes a Common Info field and a Link Info field, wherein the Link Info field includes a first Per-STA Profile subelement, wherein the first Per-STA Profile subelement includes a STA MAC Address Present subfield, wherein the STA MAC Address Present subfield indicates a presence of a STA MAC Address subfield and is set to 1 if the STA MAC Address subfield is present in a STA Info field in the first Per-STA Profile subelement, and wherein the STA MAC Address subfield of the STA Info field carries a MAC address of the second non-AP STA.
2. The method of claim 1, wherein a third AP operating on a third link is further affiliated with the AP MLD, wherein a third non-AP STA operating on the third link is further affiliated with the non-AP MLD.
3. The method of claim 2, wherein the Link Info field further includes a second Per-STA Profile subelement, wherein the second Per-STA Profile subelement includes a STA MAC Address Present subfield, wherein the STA MAC Address Present subfield indicates a presence of a STA MAC Address subfield and is set to 1 if the STA MAC Address subfield is present in a STA Info field in the second Per-STA Profile subelement, and wherein the STA MAC Address subfield of the STA Info field carries a MAC address of the third non-AP STA.
4. The method of claim 3, wherein the first link is an association link, wherein the second and third links are non-association links.
5. The method of claim 1, further comprising: receiving, by the first non-AP STA, link setting information from the first AP; transmitting, by the first non-AP STA, an association request frame to the first AP; and receiving, by the first non-AP STA, an association response frame from the first AP.
6. The method of claim 5, wherein the link setting information is included in a beacon frame or a probe response frame, wherein the link setting information includes link capabilities of the AP MLD, channel information, or information on a link set that cannot be simultaneously transmitted and received, wherein the ML element is included in the association response frame.
7. The method of claim 1, wherein the ML element further includes an element identifier field, a length field, an element identifier extension field, and a Multi-Link (ML) control field, wherein the ML control field includes a type field, wherein if the type field is set to 0, the ML element is a Basic ML element, wherein if the type field is set to 1, the ML element is a probe request ML element.
8. The method of claim 2, further comprising: transmitting, by the second non-AP STA, a first initial request frame to the second AP; receiving, by the second non-AP STA, a first initial response frame from the second AP; transmitting, by the third non-AP STA, a second initiation request frame to the third AP; and receiving, by the third non-AP STA, a second initiation response frame from the third AP; wherein the first initiation request frame and the first initiation response frame are exchanged in the second link based on the MAC address of the second non-AP STA, wherein the second initiation request frame and the second initiation response frame are exchanged in the third link based on a MAC address of the third non-AP STA.
9. The method of claim 8, wherein the first and second initiation request frames include MAC addresses of the AP and non-AP MLDs, identifiers or association Identifiers (AIDs) of the AP and non-AP MLDs.
10. A access point (AP) multi-link device (MLD) in a wireless local area network (WLAN) system,

the AP MLD comprising: a memory; a transceiver; and a processor being operatively connected to the memory and the transceiver, wherein the processor is configured to: receive, by a first AP affiliated with the AP MLD, a multi-link (ML) element from a first non-AP station (STA) affiliated with a non-AP MLD; and decode, by the first AP, the ML element, wherein the first AP operating on a first link and a second AP operating on a second link are affiliated with the AP MLD, wherein the first non-AP STA operating on the first link and a second non-AP STA operating on the second link are affiliated with the non-AP MLD, wherein the ML element includes a Common Info field and a Link Info field, wherein the Link Info field includes a first Per-STA Profile subelement, wherein the first Per-STA Profile subelement includes a STA MAC Address Present subfield, wherein the STA MAC Address Present subfield indicates a presence of a STA MAC Address subfield and is set to 1 if the STA MAC Address subfield is present in a STA Info field in the first Per-STA Profile subelement, and wherein the STA MAC Address subfield of the STA Info field carries a MAC address of the second non-AP STA.

11. A method performed by a non-access point (non-AP) multi-link device (MLD) in a wireless local area network (WLAN) system, the method comprising: generating, by a first non-AP station (STA) affiliated with the non-AP MLD, a multi-link (ML) element; and transmitting, by the first non-AP STA, the ML element to a first AP affiliated with an AP MLD, wherein the first AP operating on a first link and a second AP operating on a second link are affiliated with the AP MLD, wherein the first non-AP STA operating on the first link and a second non-AP STA operating on the second link are affiliated with the non-AP MLD, wherein the ML element includes a Common Info field and a Link Info field, wherein the Link Info field includes a first Per-STA Profile subelement, wherein the first Per-STA Profile subelement includes a STA MAC Address Present subfield, wherein the STA MAC Address Present subfield indicates a presence of a STA MAC Address subfield and is set to 1 if the STA MAC Address subfield is present in a STA Info field in the first Per-STA Profile subelement, and wherein the STA MAC Address subfield of the STA Info field carries a MAC address of the second non-AP STA.

12. The method of claim 11, wherein a third AP operating on a third link is further affiliated with the AP MLD, wherein a third non-AP STA operating on the third link is further affiliated with the non-AP MLD.

13. The method of claim 12, wherein the Link Info field further includes a second Per-STA Profile subelement, wherein the second Per-STA Profile subelement includes a STA MAC Address Present subfield, wherein the STA MAC Address Present subfield indicates a presence of a STA MAC Address subfield and is set to 1 if the STA MAC Address subfield is present in a STA Info field in the second Per-STA Profile subelement, and wherein the STA MAC Address subfield of the STA Info field carries a MAC address of the third non-AP STA.

14. The method of claim 13, wherein the first link is an association link, wherein the second and third links are non-association links.

15. The method of claim 11, further comprising: transmitting, by the first AP, link setting information to the first non-AP STA; receiving, by the first AP, an association request frame from the first non-AP STA; and transmitting, by the first AP, an association response frame to the first non-AP STA.

16. A non-access point (non-AP) multi-link device (MLD) in a wireless local area network (WLAN) system, the non-AP MLD comprising: a memory; a transceiver; and a processor being operatively connected to the memory and the transceiver, wherein the processor is configured to: generate, by a first non-AP station (STA) affiliated with the non-AP MLD, a multi-link (ML) element; and transmit, by the first non-AP STA, the ML element to a first AP affiliated with an AP MLD, wherein the first AP operating on a first link and a second AP operating on a second link are affiliated with the AP MLD, wherein the first non-AP STA operating on the first link and a second non-AP STA operating on the second link are affiliated with the non-AP MLD, wherein the ML element includes a Common Info field and a Link Info field, wherein the Link Info field includes a

first Per-STA Profile subelement, wherein the first Per-STA Profile subelement includes a STA MAC Address Present subfield, wherein the STA MAC Address Present subfield indicates a presence of a STA MAC Address subfield and is set to 1 if the STA MAC Address subfield is present in a STA Info field in the first Per-STA Profile subelement, and wherein the STA MAC Address subfield of the STA Info field carries a MAC address of the second non-AP STA.